

From shallow to deep neural networks: AI-driven models of lepton-nucleus scattering

Remarks on generalization

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September 18, 2026



Remarks on GENERALIZATION

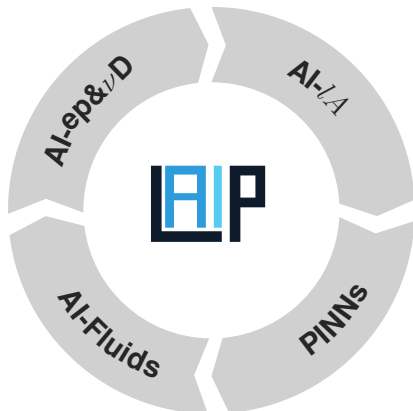
Everything that follows is one question: **when is a model right where nobody has measured yet?**

With shallow networks, we addressed it by *ranking* explicitly chosen models using evidence and the Occam factor.

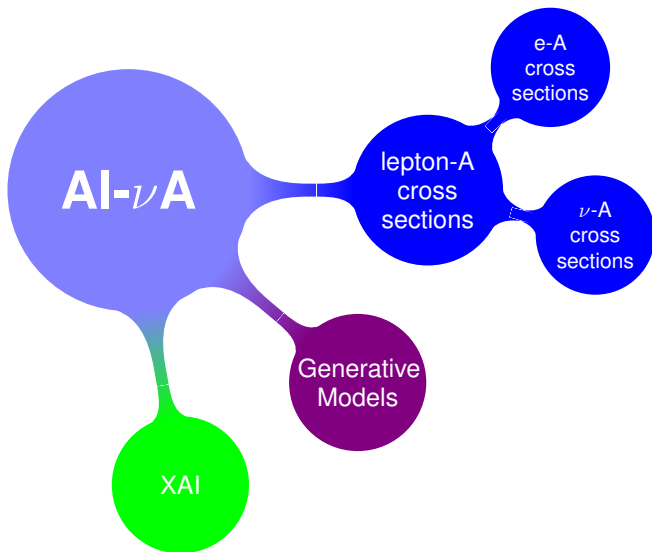
Deep networks can learn several useful representations within one overparameterized model.

The physics goal: a *model* of ℓA interactions that extrapolates — not a fit that interpolates.

- 1 LAIP and the ℓA modelling programme.
- 2 Shallow networks: evidence and model selection.
- 3 Why depth becomes necessary.
- 4 Representation learning and knowledge transfer.
- 5 Physics interpretation and open problems.



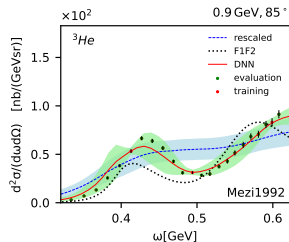
<https://kgraczyk.github.io/laip/>



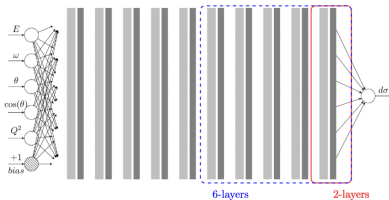
Related presentations: Luis on generative models and Beata on DNN cross-section models and preliminary XAI results.

Where we are heading

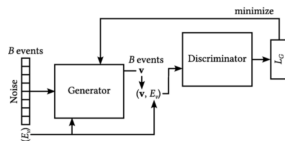
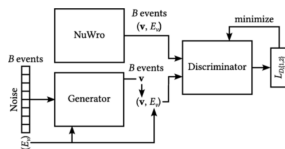
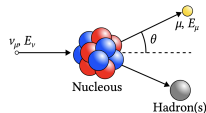
electron- ^3He scattering from electron- ^{12}C scattering
model via deep learning



Why so deep?

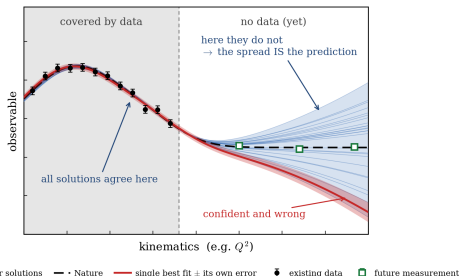


Let us build the AI- and data-driven model of ℓA interaction and try to understand what it encodes.



Let us also generate events with a deep learning model.

Why computational depth?



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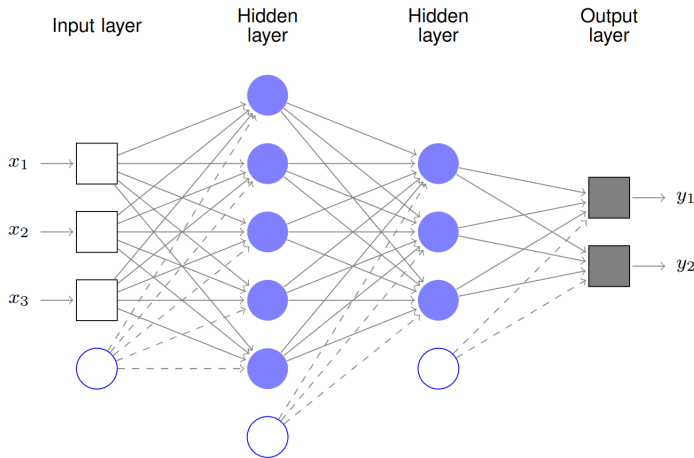
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- The goal goes beyond interpolation in the measured domain. We need representations that remain useful in new kinematic regions and for new nuclear targets.
- Computational depth lets a model build features from features and reuse them across related tasks.
- Depth and parameter count W are different notions. This distinction matters once Hessian-based model selection ceases to scale.



Multilayer Perceptron (MLP) - several hidden layers



features $\rightarrow$ transform $\rightarrow$ features' $\rightarrow$ transform $\rightarrow$ features'' $\rightarrow$ weight $\rightarrow$ **target**

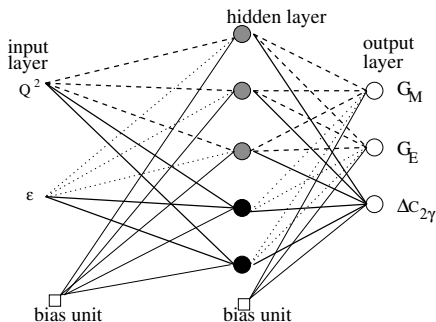
One hidden layer is enough. Completeness is not a working method.

- **How many neurons?** The theorem does not say. To be accurate *everywhere* in a d -dimensional box you have to cover it: assuming only Lipschitz smoothness, the number of hidden units needed for accuracy ε scales as $N \sim \varepsilon^{-d}$.
 $d = 1, \varepsilon = 10^{-2}: N \sim 10^2.$ $d = 3, \varepsilon = 10^{-2}: N \sim 10^6.$
 $F_A(Q^2)$ is easy; $d\sigma(E, \theta, \omega)$ is a different problem.
- **Will the training find it?** The theorem lives in function space; gradient descent lives in weight space. **Existence is not reachability.**
- **So why go deeper?** The same function, far fewer neurons: there are known examples that need $\exp(d)$ units with one hidden layer, but only $\text{poly}(d)$ with three. A deep network **builds features out of features** instead of listing them one by one.

Cybenko (1989); Hornik (1991); Leshno *et al.* (1993); Barron (1993); Eldan & Shamir (2016); Telgarsky (2016).

- An early use of neural networks in physics was the data-driven parametrization of physical quantities:
 - flexible, minimally biased functional forms;
 - complexity adapted to the information in the data;
 - nucleon form factors and two-photon exchange corrections (2010–2018), before the modern deep-learning era.

$$\mathcal{N}_{g,t} : \mathbb{R}^2 \mapsto \mathbb{R}^3, \quad \mathcal{N}_{g,t}(Q^2, \epsilon; \vec{w}) = \begin{pmatrix} G_M \\ G_E \\ \Delta C_{2\gamma} \end{pmatrix}$$

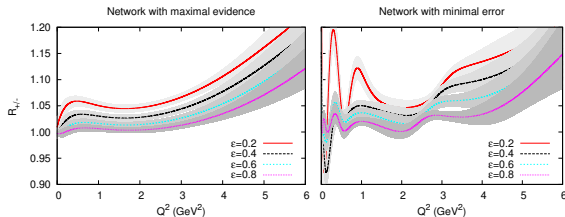


ANN: 2-(3-2)-3: $\mathcal{N}_{3,2}$

one hidden layer, two groups of units — those connected to Q^2 only and those connected to (Q^2, ϵ) : $\mathcal{N}_{3,2}$.

- **Three** unknown functions and **three** types of data.
- **The polarization-transfer ratio is relatively insensitive to TPE. but TPE must be considered in the cross section data analysis**
- Inconsistency between PT and cross-section data → extraction of the TPE correction
- $R_{+/-} = 0$ at $\epsilon = 0$ for several Q^2 values, from C -symmetry and crossing symmetry of scattering amplitude
- Parametrize form factors and TPE with neural networks.
- Before TensorFlow, PyTorch, ..., C++ library by Graczyk and Juszczak

$$R_{+/-} \approx 1 - 2\Delta_{2\gamma}, \text{ Graczyk and Juszczak, J.Phys. G42 (2015) 3, 034019}$$



$$Loss + \frac{\alpha}{2} \sum_i w_i^2 \rightarrow Loss + \frac{1}{2} \left(\alpha_h \sum_{i \in \text{hidden}} w_i^2 + \alpha_b \sum_{i \in \text{bias}} w_i^2 + \alpha_l \sum_{i \in \text{linear}} w_i^2 \right)$$

- In standard practice α is a **hyperparameter**: fixed by hand or tuned on a validation set, i.e. chosen *outside* the fit.
- The Bayesian framework promotes α to a quantity determined by the data itself — and, with it, the whole architecture.

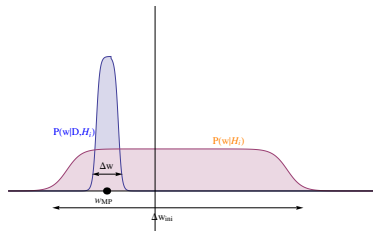
- For many problems posterior has a strong peak at $\{w_i\}_{MP} \rightarrow$ 1-dim. case, then

$$\underbrace{P(\mathcal{D}|\mathcal{N})}_{\text{Evidence}} \approx \underbrace{P(\mathcal{D}|\{w_i\}_{MP}, \mathcal{N})}_{\text{Best fit likelihood}} \underbrace{P(\{w_i\}_{MP}|\mathcal{N})\Delta w}_{\text{Occam factor}}$$

- assume that the prior has uniform distribution in wide Δw_{ini} and $P(w_{MP}|\mathcal{N}) \sim 1/\Delta w_{ini}$ then the Occam factor reads,

$$\text{Occam factor} = \frac{\Delta w}{\Delta w_0} \rightarrow \frac{V(\text{Posterior})}{V(\text{Prior})}$$

Occam factor naturally penalizes too complex models D.J.C. MacKay, Neural Computation 4, 415 (1992)



The whole method is based on Hessian approximation ...

The evidence is a W -dimensional integral over all weight configurations:

$$\mathcal{P}(\mathcal{D}|\alpha, \mathcal{N}) = \int d^W w \mathcal{P}(\mathcal{D}|\{w_i\}, \mathcal{N}) \mathcal{P}(\{w_i\}|\alpha, \mathcal{N}) = \int d^W w e^{-S(\mathcal{D}, \{w_i\})}$$

- **The only approximation:** S has a sharp minimum at $\{w_i\}_{MP}$ — expand to second order,

$$S \approx S_{MP} + \frac{1}{2} \Delta w^T A \Delta w, \quad A = H + \alpha I, \quad H_{kj} = \nabla_k \nabla_j \chi_{ex}^2.$$

Finding $\{w_i\}_{MP}$ is training the network. The integral is then done analytically:

$$\ln \mathcal{P}(\mathcal{D}|\alpha, \mathcal{N}) \approx \underbrace{-\chi_{ex}^2(\{w_i\}_{MP})}_{\text{misfit}} \underbrace{-\alpha E_w - \frac{1}{2} \ln |A| + \frac{W}{2} \ln \alpha}_{\text{Occam factor}} + \text{const}(\mathcal{D})$$

- **α is not ours to choose:** maximizing the same evidence gives a self-consistency condition, solved iteratively ($\alpha_{k+1} = \gamma(\alpha_k)/2E_w$), with

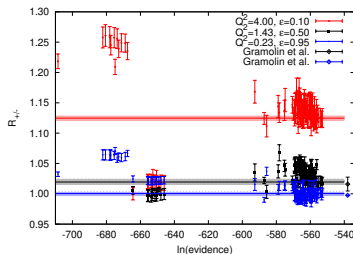
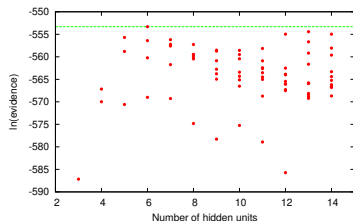
$$\gamma = \sum_{i=1}^W \frac{\lambda_i}{\lambda_i + \alpha_{MP}}, \quad \lambda_i = \text{eig}(H)$$

γ = the effective number of parameters that the data actually determine —
 $\gamma \rightarrow W$: every weight is pinned by the data; $\gamma \ll W$: most directions are fixed by the prior.

- **Now the approach is fully differentiable, and the α 's are obtained from bilevel optimization: KMG, Witkowski, Phys. Rev. E 113, 055307 (2026).**



Evidence versus architecture, and versus the observable



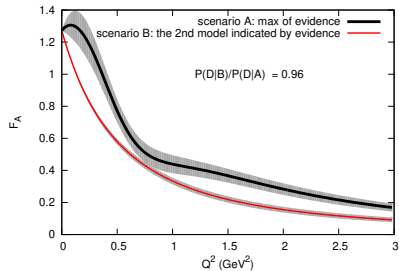
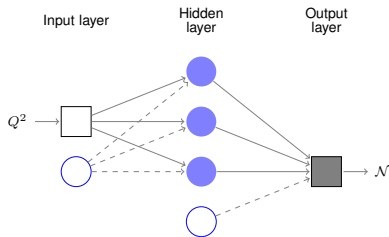
$$R_{+/-} \approx 1 - 2\Delta_{2\gamma}$$

- The spread of $\ln(\text{evidence})$ at fixed M is due to the division of the network into Q^2 -only and (Q^2, ϵ) dependent structures.
- A single dot corresponds to the best model in the family.
- Models of low evidence predict $R_{+/-}$ scattered over a wide range.
- As the evidence increases, the predictions converge — and converge onto the measured values.
- The ranking selects models that are right about an observable they were not fitted to.

Two competitive scenarios: on the edge of dipole

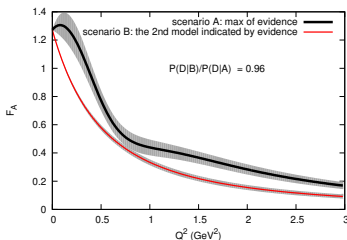
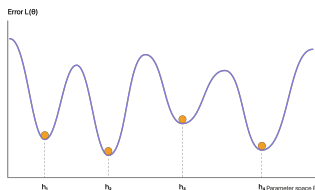
- Analyze the bubble chamber ANL ν -deuteron scattering data to obtain F_A , Phys.Rev.C 99 (2019) 2, 025204
- FA by neural network

$$F_A(Q^2) = F_A^{\text{dipole}}(Q^2) \times \mathcal{N}_M(Q^2; \{w_i\})$$



- Small corrections affect the model ranking.
- The first bin is in tension with the constraint $F_A(0) = g_A$.
- Imposing $dF_A/dQ^2|_{Q^2=0} < 0$ selects a solution close to the dipole shape.

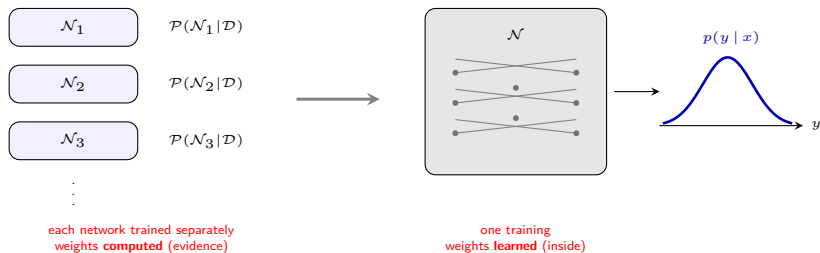
- An early example of physics-encoded neural networks (PeNNs).
- Encoding exact physical assumptions directly in the architecture led to inconclusive optimization results. **Soft physical constraints** later became a standard strategy in physics-informed neural networks.
- Reliable model selection required training thousands of candidate networks.
- The Laplace-evidence framework is practical while the Hessian remains tractable and the data constrain enough directions in weight space.
- **Several statistically competitive solutions can carry physical information.**
- The covariance-based Levenberg–Marquardt algorithm worked well for small networks but does not scale to modern deep architectures.



slide 15: $\mathcal{P}(\mathcal{D}|B)/\mathcal{P}(\mathcal{D}|A) = 0.96$

- Two physically distinct F_A solutions are **statistically indistinguishable**. A single best fit therefore does not exhaust the information supported by the data.
- Deep learning allows alternative representations to coexist within one overparameterized model. Stochastic optimization and regularization determine which representations are effectively used.
- Related mechanisms include dropout, stochastic regularization and deep ensembles. They do not replace Bayesian model averaging, but they retain part of the predictive diversity that a single fit loses.

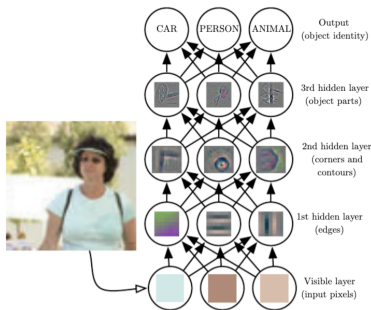
Many networks, or one network of many hypotheses



- **Shallow networks:** the model space is enumerated by hand. Each architecture is treated as a separate model classified by the evidence. Thousands of trainings, and the Hessian has to be invertible.
- **Deep networks:** alternative representations coexist within one model. Dropout can be viewed as training a large ensemble of weight-sharing subnetworks. A generative head makes the output a *conditional distribution* instead of a point. The alternatives are represented rather than listed explicitly.

Modern deep networks may contain far more parameters than constraints from the data. Their value here comes from learned representations that can be reused across related scattering problems.

- **Representation learning.** One map $(E, \theta, \omega) \mapsto d\sigma$ for a given nucleus — and the same representation reused for the next one, instead of one fit per channel.
- **Transfer of knowledge.** $e^{-12}\text{C}$ data are abundant, data on other targets are not — and the nuclear physics is largely shared. (*and, one step further, $eA \rightarrow \nu A$*)
- **Generative modelling.** NuWro needs *events*, not only $d\sigma$. A generative model samples the final state directly; a regression of the cross section does not.
- **Soft physical constraints.** Hard-encoded assumptions in the early PeNN study gave inconclusive optimization. Deep models allow constraints to enter the loss rather than the architecture, as is common in physics-informed networks.

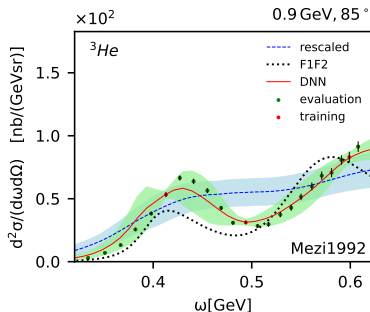


- **Why transfer is legitimate here.** The same physical mechanisms recur across nuclear targets, although their relative importance and detailed realization are target dependent. **Physics, not an ML heuristic, says the two problems share structure.**
- **Protocol.** Pre-train on inclusive $e^{-12}\text{C}$ data so that the network learns a representation of the carbon response. Fine-tune this representation for other targets: helium, lithium, oxygen, aluminium, calcium, iron.
- **Transfer learning does not reduce W ; it reduces the effective freedom that must be learned from the new data.** This is the deep-learning analogue of reducing the number of data-constrained directions, γ , in the Laplace picture of slide 13.

Shared starting point

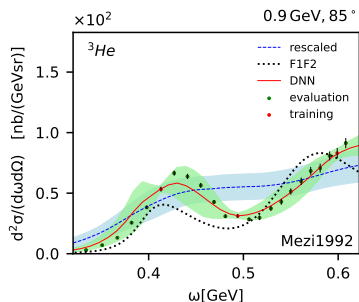
A DNN pretrained on inclusive $e^{-12}\text{C}$ data is fine-tuned for ^3He , ^6Li , ^{16}O , ^{27}Al , ^{40}Ca and ^{56}Fe .

- Shallow fine-tuning is sufficient for several medium and heavier nuclei.
 - ^3He requires adaptation through the full network.
 - The required tuning depth is therefore a physics diagnostic, not only a technical hyperparameter.
- * Phys. Rev. Lett. 135, 052502, arXiv:2609.08463



Transfer succeeds when the pretrained representation captures the shared response structure. Deeper adaptation signals a larger mismatch between the source and target nuclei.

From Carbon to Calcium



- A deep model builds its own intermediate variables. The interesting question is whether it rediscovers the ones we know — e.g. a **scaling variable**.
- The transferred representation appears to encode patterns of variation more strongly than target-specific normalization. Fix (E, θ) : $d^2\sigma/d\omega d\Omega$ is a single curve — **one trajectory**. The data are already per nucleon, so the trivial A -scaling is divided out. What is left is small and saturating: the peak broadens with k_F , shifts slightly with the removal energy, Pauli blocking cuts the low- ω side, strength moves into the dip region.
- * Beata's talk and [arxiv:2609.08463](https://arxiv.org/abs/2609.08463)

Doing physics in the deep-learning way

- Shallow networks: every model was enumerated and *weighted* using evidence and the Occam factor. It worked, and it does not scale.
- Deep networks can represent several alternatives within one model, but **a principled weighting of those alternatives at large W remains an open problem.**
- Transfer tests generalization directly: the representation learned from carbon must remain useful for a different nucleus.
- XAI asks which physical structures the surrogate has encoded.

The END.